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ELECTRICAL ASSEMBLY

Abstract

An electrical assembly, comprising an electrical contactor including a first terminal; a bus bar connected to the first terminal; and a heat spreader molded onto the electrical contactor and the bus bar. The heat spreader can comprise a thermoset material that is electrically insulating and thermally conductive. The bus bar can be welded with the first terminal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of and priority to U.S. Provisional Application 63/560,099 filed Mar. 1, 2024, the disclosure of which is hereby incorporated by reference in its entirety as though fully set forth herein.

TECHNICAL FIELD
[0002] The present disclosure generally relates to electrical assemblies, including electrical contactor assemblies that can, for example, be utilized in connection with vehicles including high voltage systems.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0003] While the claims are not limited to a specific illustration, an appreciation of various aspects may be gained through a discussion of various examples. The drawings are not necessarily to scale, and certain features may be exaggerated or hidden to better illustrate and explain an innovative aspect of an example. Further, the exemplary illustrations described herein are not exhaustive or otherwise limiting, and embodiments are not restricted to the precise form and configuration shown in the drawings or disclosed in the following detailed description. Exemplary illustrations are described in detail by referring to the drawings as follows:
[0004] FIG. **1** is a perspective view generally illustrating an embodiment of an electrical assembly according to teachings of the present disclosure.
[0005] FIG. **2** is a perspective view generally illustrating an embodiment of a contactor according to teachings of the present disclosure.
[0006] FIG. **3** is a perspective view generally illustrating an embodiment of an electrical assembly, with a heat spreader hidden, according to teachings of the present disclosure.
[0007] FIG. **4** is a cross-sectional view generally illustrating an embodiment of an electrical assembly, with a heat spreader hidden, according to teachings of the present disclosure.
[0008] FIG. **5** is a perspective view generally illustrating an embodiment of an electrical assembly according to teachings of the present disclosure.
[0009] FIG. **6** is a cross-sectional view generally illustrating an embodiment of an electrical assembly according to teachings of the present disclosure.
[0010] FIG. **7** is an enlarged version of a part of FIG. **6** along with embodiments of a cold plate and a charger according to teachings of the present disclosure.
[0011] FIG. **8** is a cross-sectional view generally illustrating an embodiment of portions of an electrical assembly and a mold according to teachings of the present disclosure.
[0012] FIG. **9** is a flow chart generally illustrating an embodiment of a method of assembling an electrical assembly according to teachings of the present disclosure.
[0013] FIG. **10** is a schematic view generally illustrating an embodiment of a vehicle according to teachings of the present disclosure.
[0014] FIG. **11** is a flow chart generally illustrating an embodiment of a method of assembling a vehicle according to teachings of the present disclosure.
[0015] FIG. **12** is a flow chart generally illustrating an embodiment of a method of operating a vehicle according to teachings of the present disclosure.
[0016] FIG. **13** is a perspective view generally illustrating an embodiment of a contactor according to teachings of the present disclosure.
[0017] FIG. **14** is a perspective view generally illustrating an embodiment of an electrical assembly according to teachings of the present disclosure.
DETAILED DESCRIPTION
[0018] Reference will now be made in detail to embodiments, examples of which are illustrated in

the accompanying drawings. In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the various described embodiments. However, it will be apparent to one of ordinary skill in the art that the various described embodiments may be practiced without these specific details. In other instances, well-known methods, procedures, components, circuits, and networks have not been described in detail so as not to unnecessarily obscure aspects of the embodiments.

[0019] Referring to FIG. **1**, an electrical assembly **20** includes a first contactor **22**, a second contactor **24**, a set of bus bars **26**, and/or a heat spreader **28**. The set of bus bars **26** includes a first bus bar **40**, a second bus bar **42**, a third bus bar **44**, and/or a fourth bus bar **46**. The first and second bus bars **40**, **42** are electrically connected to the first contactor **22**, and the first contactor **22** selectively electrically connects the first bus bar **40** with the second bus bar **42**. The third and fourth bus bars **44**, **46** are electrically connected to the second contactor **24**, and the second contactor **24** selectively electrically connects the third bus bar **44** with the fourth bus bar **46**. Operation of the contactors **22**, **24** generates heat, such as via electrical current flowing through the contactors **22**, **24** (e.g., a low voltage coil, a movable contact, terminals, etc.) and the set of bus bars **26**. The heat spreader **28** facilitates dissipating at least some of the generated heat.

[0020] Referring to FIG. **2**, the first contactor **22** includes a housing **60**, a first terminal **62**, and a second terminal **64**. The housing **60** is generally rectangular with one or more rounded edges, but can include other configurations. The first terminal **62** and the second terminal **64** extend from a first side **66** of the housing **60**. An electrical connector **58** (FIG. **1**) extends from a second side **68** of the housing **60** that is opposite the first side **66**. The electrical connector **58** (FIG. **1**) is a low voltage electrical connector **58** and is utilized to control operation of the first contactor **22** to selectively electrically connect the first terminal **62** with the second terminal **64**, which selectively electrically connects the first bus bar **40** and the second bus bar **42** (FIG. **1**). The housing **60** includes a flange **70** that extends outward generally parallel to the first side **66**. Optionally, the flange **70** defines at portion of the first side **66**. The flange **70** includes a first section **72** and a second section **74** that extends beyond the first section **72** such that the flange **70** includes a stepped configuration. The second section **74** may define the portion of the first side **66**. The first section **72** is closer to the second side **68** than the second section **74**.

[0021] The housing **60** includes a plurality of tabs **80** that extend from the first side **66** and are separated by a plurality of gaps **82**. The tabs **80** are spaced about a perimeter of the first side **66**. The plurality of tabs **80** include a plurality of first tabs **90** and a plurality of second tabs **92**. The first tabs **90** are longer and extend farther from the first side **66** than the second tabs **92**. The first tabs **90** include a first group of first tabs **94** and a second group of first tabs **96**. The second tabs **92** include a first group of second tabs **98** and a second group of second tabs **100**. The first group of first tabs **94** are spaced along a first long side **122** of the housing **60**, part of a first short side **124** of the housing **60**, and part of a second short side **126** of the housing **60**. The second group of first tabs **96** are spaced along a second long side **128** of the housing **60**. The first and second groups of second tabs **98**, **100** are spaced along the first and second short sides **124**, **126**, respectively. The first and second groups of second tabs **98**, **100** are disposed between the first and second groups of first tabs **94**, **96** to define a first recess **102** and a second recess **104**. The heights of the recesses **102**, **104** are equal to the difference in height between the first tabs **90** and the second tabs **92**. Some or all of the plurality of tabs **80** can extend from the flange **70**. The second contactor **24** can include the same or similar configuration as the first contactor **22**.

[0022] Referring to FIG. **3**, which illustrates the contactors **22**, **24** and the set of bus bars **26** with the heat spreader **28** hidden, the first bus bar **40** is disposed partially in the recess **102** and the second bus bar **42** is disposed partially in the recess **104**. For example, the first bus bar **40** extends through the recess **102** to connect with the first terminal **62** and is in contact with outer edges of the first group of second tabs **98**. Additionally or alternatively, the second bus bar **42** extends through the recess **104** to connect with the second terminal **64** and is in contact with outer edges of the second group of second tabs **100**. Referring to FIG. **4**, the heights **102**H, **104**H of the recesses **102**, **104** can be at least as large or larger than the thicknesses **40**T, **42**T of the bus bars **40**, **42**. For example, the first tabs **90** extend beyond at least the portions of the bus bars **40**, **42** disposed on the second tabs **92**.

[0023] Referring again to FIG. **3**, the first bus bar **40** is fixed to the first terminal **62**, and the second bus bar **42** is fixed to the second terminal **64**. For example, the first bus bar **40** includes an aperture **110** into which the first terminal **62** extends, and the second bus bar **42** includes an aperture **112** into which the second terminal **64** extends. The bus bars **40**, **42** are welded (e.g., laser welded) to the terminals **62**, **64**. Optionally, the outer surfaces **114**, **116** of the terminals **62**, **64** are coplanar with the outer surfaces **118**, **120** of the bus bars **40**, **42**. Additionally or alternatively, the set of bus bars **26** are connected (e.g., fixed) to the terminals **62**, **64** other than via welding, such as via fasteners. In some configurations, such as illustrated in FIG. **2**, the terminals **62**, **64** include shoulders **130**, **132**. The shoulders **130**, **132** are portions of the terminals **62**, **64** with increased outer dimensions (e.g., diameters for circular terminals) relative to at least some other portions of the terminals **62**, **64**. Some or all of the outer dimensions of the shoulders **130**, **132** are larger than the inner dimensions of the apertures **110**, **112** (FIG. **3**) such that the bus bars **40**, **42** (FIG. **3**) are disposed on the shoulders **130**, **132**, and the shoulders **130**, **132** limit the insertion depth of the terminals **62**, **64**. Optionally, the shoulders **130**, **132** are aligned with (e.g., extend beyond the first side **66** the same distance as) the second tabs **92** such that the bus bars **40**, **42** contact the shoulders **130**, **132** and the second tabs **92** and are parallel to the first side **66**.

[0024] Referring to FIGS. **1** and **5**, the heat spreader **28** is formed onto the contactors **22**, **24** and the set of bus bars **26**, such as after the set of bus bars **26** are fixed to the contactors **22**, **24**. The heat spreader **28** includes a base section **150**, a plurality of extensions **152** that extend from the base section **150**, and a plurality of fins **154** that extend from the base section **150**. The base section **150** extends over portions of the contactors **22**, **24** and portions of the set of bus bars **26**, such as to facilitate heat dissipation from the contactors **22**, **24** and/or the set of bus bars **26**. The extensions **152** cover some surfaces of the ends **162** of the set of bus bars **26**, and define open portions **160** that expose at least one surface of the ends **162** of the set of bus bars **26**, such as for electrical connection with other components (e.g., cables, wires, bus bars, fuses, terminals, among others). The ends **162** can, for example, include apertures **164** for engaging fasteners for electrically connecting the set of bus bars **26** with other components. Additionally or alternatively, the ends **162** can include clinch nuts **166** extending into and/or through the apertures **164** (FIG. **5**). The clinch nuts **166** can be connected to the ends **162** before or after the set of bus bars **26** are welded with the contactors **22**, **24**.

[0025] The base section **150** includes a plurality of mounting apertures **170** that extend through the base section **150** for mechanically mounting the electrical assembly **20** to another component. The base section **150** includes a first surface **180** from which the plurality fins **154** extend, and includes a second surface **182** opposite the first surface **180**. The second surface **182** is planar and can be formed flush with the outer edges of the first tabs **90** such that the heat spreader **28** and/or the first tabs **90** provide at least a portion of a planar mounting surface **184** of the electrical assembly **20** (FIG. **5**). The heat spreader **28** can include one or more locating holes **186** that may be formed by pins **284** of a mold **270** (FIG. **8**), such as to support and locate the set of bus bars **26** during formation of the heat spreader **28**.

[0026] Referring to FIG. **6**, the first contactor **22** includes the first and second terminals **62**, **64** and an internal assembly **190** that selectively electrically connects the first and second terminals **62**, **64**. The internal assembly **190** can include an electromagnet **192** that controls movement of an armature **194** into and out of contact with the terminals **62**, **64**. The electromagnet **192** is connected to the electrical connector **58**.

[0027] Referring to FIGS. **6** and **7**, the heat spreader **28** covers various portions of the electrical assembly **20** and forms a generally plate-like structure. The base section **150** covers the outer surfaces **114-120** of the terminals **62**, **64** and the bus bars **40**, **42**. The base section **150** includes a first portion **200** disposed between and in contact with the first bus bar **40** and the second bus bar **42**. The base section **150** includes a second portion **202** disposed between and in contact with the first bus bar **40** and the first side **66**. The second portion **202** is also in contact with the first terminal **62**, such as with a surface of the first terminal **62** that extends from the shoulder **130** to or toward the first side **66**. For example, the housing **60**, the first bus bar **40**, and the heat spreader **28** can cooperate to cover some, most, or all of the first terminal **62**. The base section **150** includes a third portion **204** disposed between and in contact with the second bus bar **42** and the first side **66**. The third portion **204** is also in contact with the second terminal **62**, such as with a surface of the first terminal **62** that extends from the shoulder **132** to or toward the first side **66**. For example, the housing **60**, the second bus bar **42**, and the heat spreader **28** can cooperate to cover some, most, or all of the second terminal **64**. Additionally or alternatively, the heat spreader **28** covers some, most, or all of the first side **66** of the housing **60**. The base section **150** includes a fourth portion **206** that covers the outer the outer surfaces **118**, **120** of the set of bus bars **26**, apart from the open portions **160**, and the outer surfaces **114**, **116** of the terminals **62**, **64**.

[0028] Referring to FIG. **7**, the base section **150** includes a fifth portion **208** disposed on the second section **74** of the flange **70** at the first side **66** of the housing **60**. The base section **150** includes a sixth portion **210** that covers a side **212** of the second section **74** that is angled (e.g., at right angle) relative to the first side **66** of the housing **60**. The base section **150** includes a seventh portion **214** that covers at least a portion of an inner surface **216** of the second section **74** of the flange **70**. The inner surface **216** is offset from the first side **66** of the housing toward the second side **68** (FIG. **6**) of the housing **60**. The seventh portion **214** helps secure the heat spreader **28** with the first contactor **22**. For example, the second section **74** is disposed at least partially between the fifth and seventh portions **208**, **214**. The first section **72** of the flange **70** includes a side **220** and an inner surface **222** offset from the inner surface **216** of the second section **74** toward the second side **68** (FIG. **6**) of the housing **60**. The heat spreader **28** (e.g., the base section **150**) may cover some, none, or all of the side **220**, and may cover some or none of the inner surface **222**. For example, the seventh portion **214** can cover all of the side **220** and none of inner surface **222** such that the seventh portion **214** is flush with the inner surface **222**. Other portions of the base section **150** can be disposed in other locations, such as on and around the plurality of tabs **80** and in the plurality of gaps **82**. The plurality of tabs **80** provide additional surface area for the base section **150** to be molded onto and can help maintain the relative positions of the contactors **22**, **24**, and the heat spreader **28**. The flange **70** can be devoid of fasteners and/or apertures for fasteners. The housing **60** can be mounted to the heat spreader **28** without fasteners.

[0029] The electrical assembly **20** can include and/or be connected to a cold plate **250**, which can dissipate heat from the heat spreader **28**, such as to further dissipate heat from the contactors **22**, **24** and/or the set of bus bars **26**. The cold plate **250** can be liquid cooled and/or air cooled. For example, the cold plate **250** can include one or more fluid passages **252** for cooling fluid, and/or can include one or more fins **254**. Optionally, another component, such as a battery charger **260** is connected to the cold plate **250**, such as opposite the electrical assembly **20**. The cold plate **250** may then operate to dissipate heat from the electrical assembly **20** and the battery charger **260**. When connecting the heat spreader **28** to the cold plate **250**, a film or layer of material (e.g., thermally conductive paste) can be applied between the heat spreader **28** and the cold plate **250** to fill gaps therebetween.

[0030] In some examples, the electrical assembly **20** may not include or be directly connected to a second heat exchanger, such as the cold plate **250**. With such examples, the second surface **182** of the heat spreader **28** can include the fins **154**, such as in addition to or instead of the first surface **180**. The configuration (e.g., number, size, geometry, arrangement) of the fins **154** can vary. In examples with the fins **154** at both of the first surface **180** and the second surface **182**, the fins **154** may provide enough cooling to sufficiently cool the contactors **22**, **24**.

[0031] The heat spreader **28** comprises a thermosetting polymer, also referred to as a thermoset, that is electrically insulating and thermally conductive. The thermoset material has a coefficient of thermal expansion that is more similar to that of the set of bus bars **26**, which may comprise copper, than other polymers. For example, the coefficient of thermal expansion of the thermoset can be within 10% of the coefficient of thermal expansion of the material of the set of bus bars **26**. Optionally, the heat spreader **28** comprises a thermoplastic, a ceramic, or both, instead of or in addition to the thermoset material.

[0032] Forming the heat spreader **28** can include molding the thermoset onto the contactors **22**, **24** and the set of bus bars **26**, such as via resin transfer molding (RTM) or compression molding, which can be conducted at lower pressures than injection molding. Referring to FIG. **8**, the molding process can include placing a first subassembly, including the first contactor **22** and the first bus bar, and a second subassembly, including the second contactor **24** and the second bus bar **42**, at least partially in a mold **270** comprising a first mold portion **272** and a second mold portion **274**. The first mold portion **272** and the second mold portion **274** are closed together to form a mold cavity **276**. A thermoset material **278** is provided to the mold cavity **276** and flows around the contactors **22**, **24** and the set of bus bars **26** in the mold cavity **276**. The thermoset material **278** can flow through one or more of the gaps **82**. Optionally, the mold **270** includes one more formations **280** (e.g., protrusions, recesses, or both), such as to form the fins **154** (FIG. **1**), the open portions **160**, and/or the mounting apertures **170** (FIG. **1**). For example, the one or more formations **280** may form sealing edges with the ends **162** to prevent the thermoset material **278** from covering the open portions **160** and/or from flowing into the apertures **164**. The one or more formations **280** can include locating pins **282**, **284** for locating one or more portions of the electrical assembly **20** during formation of the heat spreader **28**. For example, the locating pins **282** can be utilized to locate the contactors **22**, **24**, and the locating pins **284** (e.g., spring-loaded pins) can be utilized to support and locate the set of bus bars **26**. Molding the heat spreader **28** can include the thermoset material **278** flowing into and/or around the formations **280** to form the fins **154** (FIG. **1**), the open portions **160**, and/or the mounting apertures **170** (FIG. **1**). For example, the molding can include one or more formations **280** contacting the set of bus bars **26** to leave the open portions **160** uncovered, such as for electrical connection with an external component. The thermoset material **278** is allowed to cure on the contactors **22**, **24** and the set of bus bars **26**, which forms the heat spreader **28**. After curing, the assembly, including the contactors **22**, **24**, the set of bus bars **26**, and the heat spreader **28**, is removed from the mold **270**.

[0033] Referring to FIG. **9**, a method **300** of assembling the electrical assembly **20** includes connecting (e.g., welding) the set of bus bars **26** to the contactors **22**, **24** (block **302**), such as to form the first and second subassemblies. Clinch nuts **166** can be connected to the set of bus bars **26** before or after block **302**. The method **300** includes forming the heat spreader **28** onto the first and second subassemblies (block **304**), which can include molding the heat spreader **28** on the contactors **22**, **24** and the set of bus bars **26**. Optionally, forming the heat spreader **28** includes molding the heat spreader **28** on and around one or more clinch nuts **166** such that at least an outer axial surface of the one or more clinch nuts **166** is not covered by the heat spreader **28**. Optionally, the method **300** includes connecting the heat spreader **28** to a cold plate **250** (block **306**). In other configurations, the heat spreader **28** can be connected to a component that is not a cold plate and/or that does not include cooling features (e.g., a cover of a traction battery). With some examples, block **306** can be conducted in a different location than blocks **302**, **304**. For example, the electrical assembly **20**, without the cold plate **250**, may be assembled at a first location (e.g., an electronics assembly plant) and transported to a second location (e.g., a vehicle assembly plant) for assembly with a vehicle.

[0034] Referring to FIG. **10**, a schematic illustration of a vehicle **400** is provided. The vehicle **400** includes a battery **402** (e.g., a traction battery), an electric motor **404**, a battery disconnect unit (BDU) **406**, a chassis **408**, and/or the charger **260**. The BDU **406** includes the electrical assembly

20 and selectively electrically connects the battery **402** with the electric motor **404**, at least indirectly. For example, one or both of the contactors **22**, **24** selectively electrically connects the battery **402** with the electric motor **404**, such as via an inverter **412**. The inverter **412** can, for example, be attached to a housing of the electric motor **404** and be electrically connected between the BDU **406** and the electric motor **404**. The battery **402** can be a high voltage battery with a voltage of at 200V, at least 400V, less than or equal to 850V, or other values. The vehicle **400** includes the cold plate **250** and the charger **260**. The BDU **406** (e.g., the electrical assembly **20**) and the charger **260** can both be coupled with the same cold plate **250**, in some configurations. The vehicle **400** and/or the electrical assembly **20** can include a controller **410** connected to some or all of the battery **402**, the BDU **406**, the electrical assembly **20**, the charger **260**, and/or the motor **404**, such as to control, at least in part, operation thereof.

[0035] Referring to FIG. **11**, a method **500** of assembly a vehicle **400** includes assembling the electrical assembly **20** (block **502**), such as via method **300**, connecting the battery **402** with the chassis (block **504**), connecting (e.g., mechanically and thermally coupling) the electrical assembly **20** with a cold plate **250** (block **506**), connecting the cold plate **250** with the chassis **408** and/or the battery charger **260** (block **508**), and/or electrically connecting the electrical assembly **20** with the battery **402** and/or the electrical motor **404** (block **510**).

[0036] Referring to FIG. **12**, a method **600** of operating the vehicle **400** includes operating at least one of the contactors **22**, **24** to electrically connect the battery **402** with the electric motor **404** (block **602**), such as to drive the vehicle **400**. Electrical current flowing between the battery **402** and the electric motor **404** generates heat, and the method **500** includes dissipating at least a portion of the heat via the heat spreader **28** (block **604**). Dissipating heat via the heat spreader **28** can include transferring heat from one or both contactors **22**, **24** to the heat spreader **28** and/or transferring heat from the set of bus bars **26** to the heat spreader **28**. The heat spreader **28** can dissipate heat via the fins **154**. In configurations in which the electrical assembly **20** includes or is connected to a cold plate **250**, the method **500** can include the cold plate **250** dissipating heat from the heat spreader **28** (block **606**), such as heat not dissipated via the fins **154**. Optionally, the method **600** includes operating the charger **260** (block **608**), such as to charge the battery **402**, and dissipating heat from the charger **260** via the cold plate **250** (block **606**). The method **600** may be carried out, at least in part, via the controller **410**.

[0037] Referring to FIG. **13**, the housing **60** of the first contactor **22** is shown with an elongated portion **700** that connects at least two of the first tabs **90**, which can include all of the first group of first tabs **94**. The housing **60** is also shown with a second elongated portion **702** that connects at least two of the first tabs **90**, which can include all of the second group of first tabs **96**. The elongated portions **700**, **702** are disposed at distal ends of the first tabs **90** and extend across the gaps **82**. The gaps **82** allow for the thermoset material **278** to flow between the tabs **80**, such as to cover the first side **66**, portions of the set of bus bars **26**, and/or the terminals **62**, **64**. The elongated portions **700**, **702** can allow the first and second groups of first tabs **94**, **96** to function as walls instead of individual tabs. Optionally, the elongated portions **700**, **702** extend at least partially along three sides of the contactor **22** (e.g., have a C-shaped configuration) and are curved at the corners of the contactor **22**. During formation of the heat spreader **28**, the contactor **22** may be placed in the mold **270** (FIG. **8**) such that elongated portions are in contact with the mold **270** and support the contactor **22**. The mold **270** may apply a clamping force to the housing **60**, such as at the inner surface **222** of the first section **72** of the flange **70**, and the elongated portions **700**, **702** with the first tabs **90** withstand the clamping force. The housing **60** is shown with gussets **704** extending between the first side **66** of the housing **60** and the first tabs **90**, such as to strengthen the first tabs **90**. The housing **60** is also shown with locating recesses **706** (e.g., notches) that can engage locating pins of the mold **270** (FIG. **8**) during formation of the heat spreader **28**. The locating recesses **706** can be formed in the second section **74** of the flange **70**.

[0038] Referring to FIG. **14**, the set of bus bars **26** can include a variety of configurations. While shown with L-shaped configurations in other drawings, the set of bus bars **26** can include linear configurations and may be parallel with the long sides **122**, **128** of the contactors **22**, **24**.

[0039] While illustrated with two contactors **22**, **24**, the electrical assembly **20** can include other numbers of contactors, such as a single contactor or more than two contactors.

[0040] Embodiments of the electrical assembly **20** disclosed herein can have various advantages, including utilizing fewer fasteners, comprising more consistent spacing between components, improved electrical characteristics, being adaptable to various configurations and environments. Utilizing fewer fasteners can be provided by welding the bus bars with the contactor terminals and forming the heat spreader directly onto and around the bus bars and contactors (e.g., onto the flange). Utilizing fewer fasteners can also provide more consistent spacing between components, such as by reducing differences in torque applied from one fastener to the next and from one electrical assembly to the next. The more consistent spacing and/or use of a cured thermoset (e.g., a rigid material) can also provide more consistent electrical isolation between various components (e.g., bus bars, contactors), such as compared to utilizing thermal paste. Embodiments of the electrical assembly **20** can be scaled and the shape of the heat spreader can be modified to accommodate different applications and bus bar shapes, such as without or with limited further design validation.

[0041] The instant disclosure includes the following non-limiting embodiments:

[0042] An electrical assembly, comprising: an electrical contactor including a first terminal; a bus bar connected to the first terminal; and a heat spreader molded onto the electrical contactor and the bus bar.

[0043] The electrical assembly of any preceding embodiment, wherein the heat spreader comprises a thermoset material that is electrically insulating and thermally conductive.

[0044] The electrical assembly of any preceding embodiment, wherein the heat spreader comprises a ceramic material that is electrically insulating and thermally conductive.

[0045] The electrical assembly of any preceding embodiment, wherein the bus bar is mechanically and electrically connected with the first terminal.

[0046] The electrical assembly of any preceding embodiment, wherein the bus bar is welded with the first terminal.

[0047] The electrical assembly of any preceding embodiment, wherein the electrical contactor includes a housing comprising a plurality of tabs separated by a plurality of gaps.

[0048] The electrical assembly of any preceding embodiment, wherein the plurality of tabs are embedded in the heat spreader.

[0049] The electrical assembly of any preceding embodiment, wherein the bus bar is disposed partially in a recess defined at least partially by the plurality of tabs.

[0050] The electrical assembly of any preceding embodiment, wherein the housing includes a flange comprising a first section and a second section that extends beyond the first section.

[0051] The electrical assembly of any preceding embodiment, wherein the heat spreader is molded onto the second section to mechanically couple the electrical contactor with the heat spreader.

[0052] The electrical assembly of any preceding embodiment, wherein the heat spreader is molded onto the second section such that the second section is at least partially disposed between portions of the heat spreader and such that at least some of the first section is not covered by the heat spreader.

[0053] The electrical assembly of any preceding embodiment, wherein the heat spreader includes a plurality of cooling fins.

[0054] The electrical assembly of any preceding embodiment, comprising a second bus bar; wherein the electrical contactor includes a second terminal; the bus bar is welded to the first terminal; the second bus bar is welded to the second terminal; and the heat spreader is molded onto the second bus bar.

[0055] The electrical assembly of any preceding embodiment, comprising a second electrical contactor including a second contactor first terminal and a second contactor second terminal, a third bus bar welded to the second contactor first terminal, and fourth bus bar welded to the second contactor second terminal; wherein the heat spreader is molded onto the second electrical contactor, the third bus bar, and the fourth bus bar.

[0056] The electrical assembly of any preceding embodiment, comprising a second bus bar including a second bus bar first end; wherein the electrical contactor includes a housing and a second terminal; the first terminal and the second terminal extend from a first side of the housing; the second bus bar first end is welded to the second terminal; the bus bar includes a first bus bar first end welded to the first terminal; and a portion of the heat spreader is disposed between the first bus bar first end and the second bus bar first end.

[0057] The electrical assembly of any preceding embodiment, wherein the contactor housing includes a second side opposite the first side and includes an electrical connector disposed at the second side.

[0058] The electrical assembly of any preceding embodiment, wherein the electrical contactor is disposed at a first side of the heat spreader; and a second side of the heat spreader is planar.

[0059] The electrical assembly of any preceding embodiment, comprising a cold plate coupled with the second side of the heat spreader.

[0060] A vehicle, comprising: a battery; an electric motor; and a battery disconnect unit (BDU) selectively electrically connecting the battery with the electric motor, the BDU including the electrical assembly of any preceding embodiment.

[0061] A method of assembling the electrical assembly of any preceding embodiment, the method comprising: connecting the bus bar to the first terminal; and molding the heat spreader onto the electrical contactor and the bus bar.

[0062] The method of any preceding embodiment, wherein the molding includes transfer molding.

[0063] The method of any preceding embodiment, wherein the molding includes disposing the bus bar at least partially in a first mold and disposing the contactor at least partially in a second mold.

[0064] The method of any preceding embodiment, wherein the molding includes providing a thermoset material into the first mold and the second mold; and the thermoset material is electrically insulating and thermally conductive.

[0065] The method of any preceding embodiment, wherein the molding includes forming a plurality of cooling fins.

[0066] The method of any preceding embodiment, wherein the molding includes forming a plurality of mounting apertures.

[0067] The method of any preceding embodiment, wherein the molding includes leaving a portion of the bus bar uncovered for connection with an external component.

[0068] The method of any preceding embodiment, wherein the electrical contactor includes a housing comprising a plurality of tabs separated by a plurality of gaps; and

[0069] the molding includes the thermoset material flowing through at least one gap of the plurality of gaps.

[0070] A method of assembling a vehicle comprising the electrical assembly of any preceding embodiment, the method comprising: assembling the electrical assembly; connecting a battery of the vehicle with a chassis of the vehicle; connecting the electrical assembly with a cold plate; connecting the cold plate with the chassis and/or a battery charger; and electrically connecting the electrical assembly with the battery.

[0071] A vehicle including the electrical assembly of any preceding embodiment.

[0072] An electronic controller configured to implement the method of any preceding embodiment.

[0073] A non-transitory computer-readable storage medium having a computer program encoded thereon for implementing the method of any preceding embodiment.

[0074] In examples, a controller (e.g., the electronic controller **410**) may include an electronic controller and/or include an electronic processor, such as a programmable microprocessor and/or microcontroller. In embodiments, a controller may include, for example, an application specific integrated circuit (ASIC) and/or an embedded controller. A controller may include a central processing unit (CPU), a memory (e.g., a non-transitory computer-readable storage medium), and/or an input/output (I/O) interface. A controller may be configured to perform various functions, including those described in greater detail herein, with appropriate programming instructions and/or code embodied in software, hardware, and/or other medium. In embodiments, a controller may include a plurality of controllers. In embodiments, a controller may be connected to a display, such as a touchscreen display.

[0075] Various examples/embodiments are described herein for various apparatuses, systems, and/or methods. Numerous specific details are set forth to provide a thorough understanding of the overall structure, function, manufacture, and use of the examples/embodiments as described in the specification and illustrated in the accompanying drawings. It will be understood by those skilled in the art, however, that the examples/embodiments may be practiced without such specific details. In other instances, well-known operations, components, and elements have not been described in detail so as not to obscure the examples/embodiments described in the specification. Those of ordinary skill in the art will understand that the examples/embodiments described and illustrated herein are non-limiting examples, and thus it can be appreciated that the specific structural and functional details disclosed herein may be representative and do not necessarily limit the scope of the embodiments.

[0076] Reference throughout the specification to "examples, "in examples," "with examples," "various embodiments," "with embodiments," "in embodiments," "an embodiment," "with some configurations," "in some configurations," or the like, means that a particular feature, structure, or characteristic described in connection with the example/embodiment is included in at least one embodiment. Thus, appearances of the phrases "examples, "in examples," "with examples," "in various embodiments," "with embodiments," "in embodiments," "an embodiment," "with some configurations," "in some configurations," or the like, in places throughout the specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, and/or characteristics may be combined in any suitable manner in one or more examples/embodiments. Thus, the particular features, structures, or characteristics illustrated or described in connection with one embodiment/example may be combined, in whole or in part, with the features, structures, functions, and/or characteristics of one or more other embodiments/examples without limitation given that such combination is not illogical or non-functional. Moreover, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the scope thereof. The word "exemplary" is used herein to mean "serving as a non-limiting example."

[0077] It should be understood that references to a single element are not necessarily so limited and may include one or more of such element, unless the context clearly indicates otherwise. Any directional references (e.g., plus, minus, upper, lower, upward, downward, left, right, leftward, rightward, top, bottom, above, below, vertical, horizontal, clockwise, and counterclockwise) are only used for identification purposes to aid the reader's understanding of the present disclosure, and do not create limitations, particularly as to the position, orientation, or use of examples/embodiments.

[0078] "One or more" includes a function being performed by one element, a function being performed by more than one element, e.g., in a distributed fashion, several functions being performed by one element, several functions being performed by several elements, or any combination of the above. The term "at least one of" in the context of, e.g., "at least one of A, B, and C" or "at least one of A, B, or C" includes only A, only B, only C, or any combination or subset of A, B, and C, including any combination or subset of one or a plurality of A, one or a plurality of B, and one or a plurality of C. A "set" of elements can include any number of one or more elements.

[0079] Although the terms first, second, etc. are, in some instances, used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the various described embodiments. The first element and the second element are both elements, but they are not the same element.
[0080] The terminology used in the description of the various described embodiments herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in the description of the various described embodiments and the appended claims, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. The term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. Uses of “and” and “or” are to be construed broadly (e.g., to be treated as “and/or”). For example and without limitation, uses of “and” do not necessarily require all elements or features listed, and uses of “or” are inclusive unless such a construction would be illogical. The terms “includes.” “including.” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.
[0081] Joinder references (e.g., attached, coupled, connected, and the like) are to be construed broadly and may include intermediate members between a connection of elements, relative movement between elements, direct connections, indirect connections, fixed connections, movable connections, operative connections, indirect contact, and/or direct contact. As such, joinder references do not necessarily imply that two elements are directly connected/coupled and in fixed relation to each other. Connections of electrical components, if any, may include mechanical connections, electrical connections, wired connections, and/or wireless connections, among others. Uses of “e.g.” and “such as” in the specification are to be construed broadly and are used to provide non-limiting examples of embodiments of the disclosure, and the disclosure is not limited to such examples.
[0082] While processes, systems, and methods may be described herein in connection with one or more steps in a particular sequence, such methods may be practiced with the steps in a different order, with certain steps performed simultaneously, with additional steps, and/or with certain described steps omitted.
[0083] As used herein, the term “if” is, optionally, construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” is, optionally, construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context.
[0084] References to a vehicle can include one or more of a variety of vehicles, including, without limitation, a passenger car (e.g., a sedan, a pickup truck, a sport utility vehicle, a crossover, etc.), a truck, a bus, a plane, or a boat, among others.
[0085] All matter contained in the above description or shown in the accompanying drawings shall be interpreted as illustrative only and not limiting. Changes in detail or structure may be made without departing from the present disclosure.
[0086] A controller, an electronic control unit (ECU), a system, and/or a processor as described herein may include a conventional processing apparatus known in the art, which may be capable of executing preprogrammed instructions stored in an associated memory, all performing in accordance with the functionality described herein. To the extent that the methods described herein are embodied in software, the resulting software can be stored in an associated memory and can also constitute means for performing such methods. Such a system or processor may further be of

the type having ROM, RAM, RAM and ROM, and/or a combination of non-volatile and volatile memory so that any software may be stored and yet allow storage and processing of dynamically produced data and/or signals.
[0087] An article of manufacture in accordance with this disclosure may include a non-transitory computer-readable storage medium having a computer program encoded thereon for implementing logic and other functionality described herein. The computer program may include code to perform one or more of the methods disclosed herein. Such embodiments may be configured to execute via one or more processors, such as multiple processors that are integrated into a single system or are distributed over and connected together through a communications network, and the communications network may be wired and/or wireless. Code for implementing one or more of the features described in connection with one or more embodiments may, when executed by a processor, cause a plurality of transistors to change from a first state to a second state. A specific pattern of change (e.g., which transistors change state and which transistors do not), may be dictated, at least partially, by the logic and/or code.

Claims

1. An electrical assembly, comprising: an electrical contactor including a first terminal; a bus bar connected to the first terminal; and a heat spreader molded onto the electrical contactor and the bus bar.
2. The electrical assembly of claim 1, wherein the heat spreader comprises a thermoset material that is electrically insulating and thermally conductive.
3. The electrical assembly of claim 1, wherein the bus bar is welded with the first terminal.
4. The electrical assembly of claim 1, wherein the electrical contactor includes a housing comprising a plurality of tabs separated by a plurality of gaps.
5. The electrical assembly of claim 4, wherein the plurality of tabs are embedded in the heat spreader.
6. The electrical assembly of claim 4, wherein the bus bar is disposed partially in a recess defined at least partially by the plurality of tabs.
7. The electrical assembly of claim 4, wherein the housing includes a flange comprising a first section and a second section that extends beyond the first section.
8. The electrical assembly of claim 7, wherein the heat spreader is molded onto the second section to mechanically couple the electrical contactor with the heat spreader.
9. The electrical assembly of claim 8, wherein the heat spreader is molded onto the second section such that the second section is at least partially disposed between portions of the heat spreader and such that at least some of the first section is not covered by the heat spreader.
10. The electrical assembly of claim 1, wherein the heat spreader includes a plurality of cooling fins.
11. The electrical assembly of claim 1, comprising a second bus bar; wherein the electrical contactor includes a second terminal; the bus bar is welded to the first terminal; the second bus bar is welded to the second terminal; and the heat spreader is molded onto the second bus bar.
12. The electrical assembly of claim 11, comprising a second electrical contactor including a second contactor first terminal and a second contactor second terminal, a third bus bar welded to the second contactor first terminal, and fourth bus bar welded to the second contactor second terminal; wherein the heat spreader is molded onto the second electrical contactor, the third bus bar, and the fourth bus bar.
13. The electrical assembly of claim 1, comprising a second bus bar including a second bus bar first end; wherein the electrical contactor includes a housing and a second terminal; the first terminal and the second terminal extend from a first side of the housing; the second bus bar first end is welded to the second terminal; the bus bar includes a first bus bar first end welded to the first

terminal; and a portion of the heat spreader is disposed between the first bus bar first end and the second bus bar first end.

14. The electrical assembly of claim 13, wherein the housing includes a second side opposite the first side and includes an electrical connector disposed at the second side.

15. The electrical assembly of claim 1, wherein the electrical contactor is disposed at a first side of the heat spreader; and a second side of the heat spreader is planar.

16. The electrical assembly of claim 15, comprising a cold plate coupled with the second side of the heat spreader.

17. A vehicle, comprising: a battery; an electric motor; and a battery disconnect unit (BDU) selectively electrically connecting the battery with the electric motor, the BDU including the electrical assembly of claim 1.

18. A method of assembling the electrical assembly of claim 1, the method comprising: connecting the bus bar to the first terminal; and molding the heat spreader onto the electrical contactor and the bus bar.

19. The method of claim 18, wherein the molding includes transfer molding.

20. The method of claim 19, wherein the molding includes disposing the bus bar at least partially in a first mold and disposing the contactor at least partially in a second mold; the molding includes providing a thermoset material into the first mold and the second mold; the thermoset material is electrically insulating and thermally conductive; the molding includes forming a plurality of cooling fins; the molding includes forming a plurality of mounting apertures; the molding includes leaving a portion of the bus bar uncovered for connection with an external component; the electrical contactor includes a housing comprising a plurality of tabs separated by a plurality of gaps; and the molding includes the thermoset material flowing through at least one gap of the plurality of gaps.